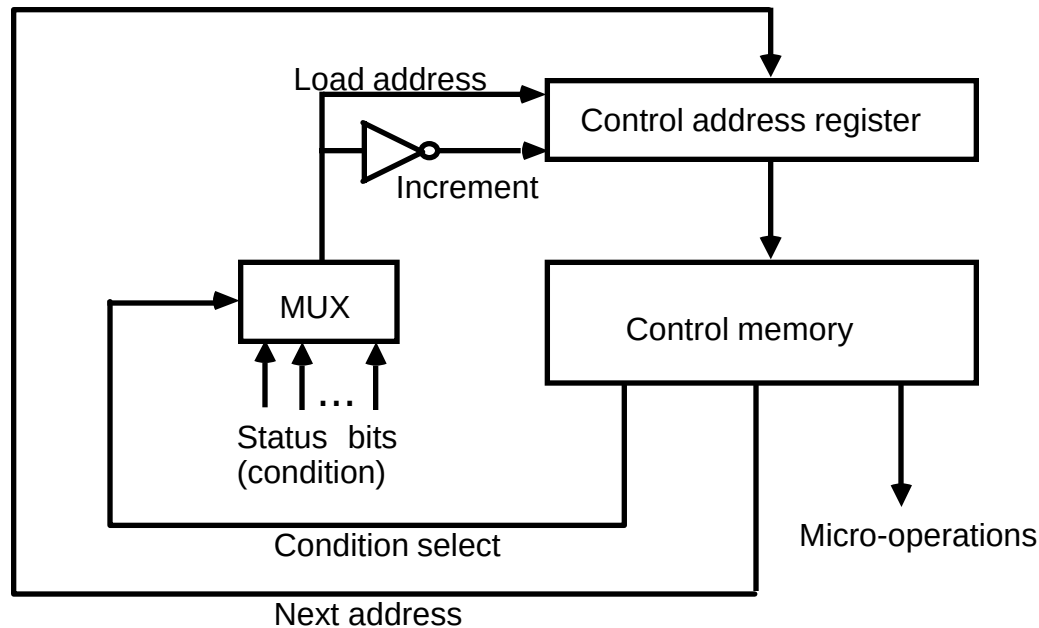


CONDITIONAL BRANCH

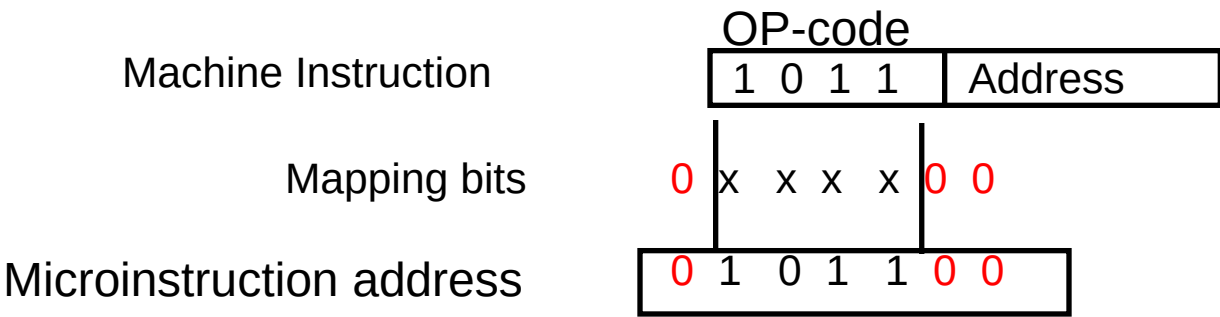


- **Conditional Branch.**
 - If *Condition* is true, then *Branch* (address from the next address field of the current microinstruction) else *Fall Through*.
 - Conditions to Test: O(overflow), N(negative), Z(zero), C(carry), etc.
- **Unconditional Branch.**
 - Fixing the value of one status bit at the input of the multiplexer to 1

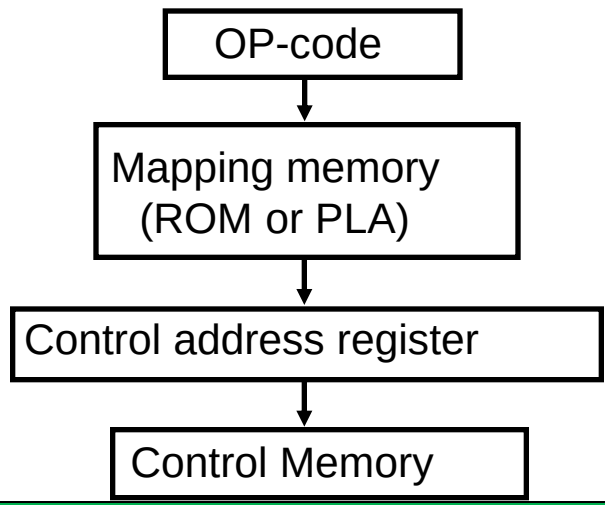


MAPPING OF INSTRUCTIONS TO MICROROUTINES

- Mapping Process: The process of converting *opcode* to *control memory address*.
- Mapping from the OP-code of an instruction to the address of the Microinstruction which is the starting microinstruction of its execution microprogram



Mapping function implemented by ROM or PLA



Mapping

- Place a “0” in the most significant bit of the address
- Transfer 4-bit Operation code bits.
- Clear the two least significant bits of the Control Address.
- Mapping Function : Implemented by *Mapping ROM* or *PLD*.
- Control Memory Size : 128 words.

